

VARUN SHARMA

AI Engineer | Technical Lead | Backend & Cloud Architecture

varunsharma64@gmail.com • +91 93060 77588 • New Delhi, India

varundev.space • github.com/varunweb6 • linkedin.com/in/varun-sharma-155a17225

SUMMARY

AI and platform engineer who designs and ships production LLM systems: multi-tenant SaaS backends, agentic workflows, RAG pipelines and AutoML services. Currently leading a delivery team at NerveSparks, owning architecture, code review and production rollout for enterprise AI products across AWS, GCP, Azure and DigitalOcean. Comfortable across the full lifecycle: FastAPI and Node services, Docker and Kubernetes, Terraform and CI/CD, plus the evaluation and observability layer that keeps LLM features reliable in production.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), SQL, Bash

AI / LLM: LangChain, LangGraph, DSPy, LiteLLM, Model Context Protocol (MCP), multi-agent orchestration, RAG (hybrid + semantic search), prompt engineering, LLM evaluation, Langfuse

Models: OpenAI GPT, Anthropic Claude, Google Gemini, Llama, Qwen, Mistral (open-weight serving via vLLM / Ollama)

Machine Learning: scikit-learn, XGBoost, SHAP explainability, feature engineering, AutoML, forecasting, classification, clustering, anomaly detection

Backend: FastAPI, Node.js, Express, Django, REST, GraphQL, WebSockets, microservices, Celery, RabbitMQ

Frontend: React, Next.js, SvelteKit, Tailwind CSS, Material UI, Puppeteer

Data Stores: PostgreSQL, MongoDB, Redis, Firestore; Weaviate, Qdrant, ChromaDB, FAISS (vector)

Cloud & DevOps: GCP (Cloud Run, Cloud Functions), AWS (EC2, S3, Lambda, ECS), Azure, DigitalOcean, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Nginx, Linux

Observability: Prometheus, Grafana, OpenTelemetry, Langfuse, ELK

Architecture: Distributed systems, multi-tenant SaaS, event-driven design, RBAC/ABAC, API design, caching strategy

EXPERIENCE

NerveSparks Pvt. Ltd.

New Delhi, India

Technical Lead — AI Engineering (promoted from Software Engineer)

Nov 2024 – Present

- Lead a cross-functional engineering team delivering enterprise AI products to multiple concurrent clients; own architecture decisions, code review standards, sprint planning and production release sign-off.
- Architected a multi-tenant enterprise AI platform with hierarchical RBAC (Platform → Organisation → Department → User) on Google Identity Platform, with tenant-level data isolation and audit trails.
- Designed agentic AI workflows using LangGraph and custom MCP servers, enabling tool-calling agents to orchestrate retrieval, external APIs and multi-step reasoning behind a single service interface.
- Built a centralised RAG service on FastAPI, LangChain and Weaviate with a provider-agnostic LLM abstraction (OpenAI, Claude, Gemini, open-weight models), reused across several client applications.
- Led an AutoML platform supporting forecasting, classification, regression, clustering and anomaly detection, with automated feature engineering, model selection and SHAP-based explainability.
- Built configurable ML data pipelines covering profiling, preprocessing and training runs, with approval workflows before models reach production.
- Shipped 15+ FastAPI microservices to Cloud Run and Kubernetes with autoscaling, cold-start tuning and zero-downtime deploys.
- Standardised infrastructure with Terraform across AWS, GCP, Azure and DigitalOcean, with Nginx load balancing fronting 10+ services.
- Cut deployment time ~60% by rebuilding CI/CD on GitHub Actions and Jenkins with automated test, build and rollback stages.
- Reduced P95 API latency ~40% via query profiling, indexing and a three-tier cache (in-memory → Redis → CDN) holding an ~85% hit ratio.
- Engineered a meeting-agent service using Puppeteer to join Google Meet, Teams and Zoom, capture audio in real time and produce searchable LLM summaries stored as semantic memory.
- Built a real-time conversational AI avatar combining speech recognition, LLM reasoning and speech synthesis for customer-facing interaction.
- Established the observability stack (Prometheus, Grafana, OpenTelemetry, Langfuse) covering both infrastructure metrics and LLM traces/cost, cutting MTTR ~50%.
- Mentor junior engineers through architecture reviews, pairing and technical planning; act as technical point of contact in client discussions.

Manav Rachna University

Web Developer Intern

Faridabad, India

Aug 2024 – Oct 2024

- Built the Student Welfare Department portal, centralising information access for 5,000+ students and faculty.
- Developed real-time feedback modules for student–faculty communication, and supported data analysis and visualisation for an HRV biomedical research project.

SELECTED PROJECTS

Enterprise AI Platform | *FastAPI · LangGraph · MCP · Weaviate · PostgreSQL · Redis · Docker · Cloud Run*

- Unified abstraction layer over multiple LLM providers with per-tenant API key management, routing and cost tracking.
- Long-term agent memory built on vector retrieval; evaluation pipelines for retrieval quality and hallucination detection reached 92% accuracy on internal test sets.

AutoML & ML Pipeline Service | *Python · FastAPI · scikit-learn · XGBoost · SHAP · PostgreSQL*

- Config-driven pipeline handling profiling, feature engineering, model selection and deployment for tabular workloads.
- SHAP explainability surfaced per-prediction drivers so non-technical stakeholders could sign off on model behaviour.

AI Document Processing System | *Python · LangChain · DSPy · ChromaDB · FastAPI*

- Modular ingestion for PDF, DOCX and image sources with semantic chunking and metadata-aware retrieval.
- DSPy-based automatic prompt optimisation improved answer relevance ~35% over hand-written prompts.

Imaginfy — AI Image Generation Platform | *React · Node.js · MongoDB · Redis · Hugging Face · Stripe*

- Text-to-image product on Stable Diffusion with credit-based Stripe billing, Redis caching and a queue for concurrent generation.

EDUCATION

Manav Rachna International University

Master of Computer Applications — GPA 8.1 / 10

Faridabad, India

Jul 2023 – Jun 2025

Maharshi Dayanand University

Bachelor of Computer Applications — GPA 7.2 / 10

Rohtak, India

Jul 2020 – May 2023